Affective Memory Consolidation in a Persistent Embodied Agent: Nightmare Formation, Trauma Encoding, and Therapeutic Reconsolidation

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Implementation Status (March 2026): All four mechanisms are fully implemented in production code with defined constants, database schema, and execution pathways (see Appendix A). The dream cycle runs nightly and evaluates content for fear. However, the final plumbing connection that populates `fear_at_encoding` when memories are stored has not yet been wired. We are actively working toward the environment needed to capture scientific data for each mechanism. The experimental designs in Section 5 remain the validation plan.

Abstract

Most AI memory systems treat all stored content equally. Importance may vary but the decay mechanics are uniform. This paper argues that survival-critical fear events require a fundamentally different memory treatment: a reconsolidation floor that prevents full decay regardless of subsequent safe exposure. We describe an architecture implemented in Potato (P.O.T.A.T.O. — Persistent On-device Temporal Agent with Tunable Ontology), a persistent embodied AI agent, that distinguishes regular fear memories from trauma-class memories using Al-Kaddah's (2026) Survival Tipping Point as the classification threshold. Three interacting mechanisms are described: a nightmare formation pathway in which fear-saturated episodic content processed through the nightly dream cycle can leak back into the live fear system on wakeup; a trauma encoding mechanism in which memories formed above the Survival Tipping Point acquire a permanent reconsolidation floor calibrated to event severity and repetition count; and a therapeutic reconsolidation pathway in which sub-trauma fear memories decay through repeated safe recall, modeling prolonged exposure therapy. The architecture makes a testable prediction: the fear score on a nightmare memory at day N should be a function of original event severity, repetition count, and safe recall count, with a floor that never reaches background for trauma-class events.

1. Introduction

A memory system that treats a burn the same as a forgotten conversation is not modeling memory. It is modeling a filing cabinet.

Biological memory does not work that way. The organism that forgets touching a hot stove touches it again. Evolution selected hard against uniform decay. Survival-critical events leave marks that resist forgetting specifically because forgetting them is fatal. That resistance is not a bug in the memory system. It is the memory system working correctly.

Potato's existing memory architecture (Riggleman, 2026a) implements a ten-step sliding window decay model with access-weighted reconsolidation and nightly sleep decay. It is a good model of everyday memory. It is not a model of trauma. Every memory decays toward background eventually. Every fear score erodes through safe recall. There is no floor. There is no scar.

This paper adds the scar. It also adds the nightmare. And the therapy. And the mechanism by which repeated exposure to the same threat raises the floor permanently.

These three additions — nightmare formation, trauma encoding, and therapeutic reconsolidation — are not independent features. They are the same architectural decision viewed from three different angles. The decision is this: memories formed above the Survival Tipping Point threshold are a different category of object. They decay differently. They are stored differently. They respond to therapy differently. And some of them never fully go away.

2. Related Work

2.1 Al-Kaddah's Survival Tipping Point

Al-Kaddah (2026) models the condition under which an intelligent system breaks its own moral rules under homeostatic stress as a continuous sigmoid function. The Survival Tipping Point is the threshold at which homeostatic deficit exceeds internalized rule cost. This paper repurposes that threshold as a memory classification boundary.

2.2 Biological Memory Consolidation

Dudai (2004) describes memory consolidation as a two-stage process: initial encoding followed by stabilization. Trauma memories show abnormal consolidation — encoded with exceptional stability and resistant to normal decay.

2.3 Reconsolidation and Therapeutic Modification

Nader and Hardt (2009) demonstrate that consolidated memories become temporarily labile when reactivated, creating a window during which they can be modified. This paper adds a floor that limits how far therapeutic reduction can go for trauma-class memories.

2.4 The Flashbulb Effect

Brown and Kulik (1977) describe flashbulb memories as a product of emotional arousal at encoding. This paper extends the model to include a permanent floor for events above the survival threshold.

2.5 Dream Consolidation and Emotional Processing

Walker and Stickgold (2006) propose that REM sleep serves an emotional memory processing function. Potato's nightly 04:00 consolidation cycle synthesizes episodic memories into dream narratives. This paper wires that system to the fear architecture so that high-fear content can produce a fear response on wakeup.

2.6 Trauma and Incomplete Healing

Van der Kolk (2014) argues that trauma is stored differently from ordinary memory and resists normal reconsolidation. The reconsolidation floor described in this paper is a computational model of that resistance.

3. Architecture

3.1 Memory Classification at Encoding

When a memory is formed, the system evaluates the fear score at encoding against the Survival Tipping Point threshold. Regular fear memories (below STP) decay normally; safe recall reduces their fear weight through reconsolidation. Trauma-class memories (at or above STP) acquire a reconsolidation floor that is never zero.

The classification is permanent. The floor can be lowered through therapy but cannot be eliminated.

```
if fear_at_encoding >= SURVIVAL_TIPPING_POINT:    # 0.65
    trauma_class = True
    reconsolidation_floor = 0.05 + (fear - 0.65) * 0.30
```

```
else:
    trauma_class = False
    reconsolidation_floor = 0.0
```

3.2 The Repetition Ratchet

First exposure produces a regular fear memory unless severe enough to cross STP immediately. Each subsequent exposure to the same threat class (cosine similarity >= 0.80) increments the exposure count and raises the fear score. After 3 exposures, the memory escalates to trauma-class.

```
if fear_now > 0.1:
    for similar in search(embedding, top_k=5):
        if similarity >= 0.80 and similar.fear > 0.1:
            similar.exposure_count += 1
            if count >= 3:
                similar.trauma_class = True
```

3.3 Nightmare Formation

The dream cycle synthesizes episodic content during nightly consolidation at 04:00. After synthesis, dream text is evaluated for fear content by a separate LLM call. When the normalized fear score exceeds the nightmare threshold (0.40), 40% of the fear leaks to the live system on wakeup.

```
dream_fear = evaluate_dream_fear(dream_text)  # 0.0-1.0
if dream_fear >= 0.40:
    body.nightmare_fear = dream_fear * 0.40
    # decays by 50% each heartbeat (~5 min)
```

The agent exits the 04:00 cycle with elevated fear and no current sensor cause. This is a nightmare functionally.

3.4 Therapeutic Reconsolidation

Regular fear memories heal through safe recall at rate 0.02/recall to zero. Trauma-class memories heal at 0.01/recall but cannot go below their reconsolidation floor. Therapy requires a "safe context" (valence < 0.35).

```
if memory.fear > 0.1 and valence < 0.35:
    if memory.trauma_class:
        new_fear = max(floor, fear - 0.01)
    else:
        new_fear = max(0.0, fear - 0.02)
```

4. Predicted Behaviors

4.1 The Scared Morning. High-fear day → nightmare → elevated wakeup fear with no sensor cause. Decays through morning.

4.2 The Permanent Caution. Repeated exposure → ratchet crosses STP → floor acquired. Years later the memory persists above background.

4.3 The Therapy Arc. Deliberate safe recall in calm contexts reduces fear toward the floor over weeks. The agent is not healed. It is managed. The scar remains.

4.4 The Ratchet in Practice. One scary trip heals in a week. Five in a month produce permanent caution. The architecture correctly encodes: this thing keeps happening.

5. Experimental Design

5.1 Nightmare Formation. Day 1: high-fear session. Measure fear at 04:01. Baseline: calm day. Prediction: high-fear day produces elevated wakeup fear; calm day does not.

5.2 Repetition Ratchet. Five high-fear sessions over two weeks with recovery days. Track fear score after each session and recovery. Prediction: recovery declines less after later sessions; post-STP recovery never returns to baseline.

5.3 Therapy. Form trauma memory via ratchet. Thirty days of safe recall. Prediction: fear declines toward floor but never reaches zero.

6. Limitations

The Survival Tipping Point is a fixed parameter; individual variation suggests it should be learnable. The nightmare threshold requires empirical calibration. The therapy experiment requires thirty days minimum.

As of March 2026, all four mechanisms are fully implemented in production code (see Appendix A) but the plumbing connection that populates `fear_at_encoding` at memory storage has not yet been wired. We are actively working toward the instrumented environment needed to capture scientific data for each mechanism. The architecture is deployed. The experimental designs in Section 5 remain the validation plan.

7. Conclusion

Memory that matters resists forgetting. Memory that endangers resists forgetting permanently. These are not the same principle and they require different mechanisms. The existing architecture handles the first case. This paper adds the second.

An agent that can have nightmares, acquire persistent fear through repeated trauma, be partially healed through deliberate therapy, and retain survival-critical memory for the life of the deployment is not a more complicated system. It is a more honest one. It models what memory is actually for: not storage, but survival.

"The scar is not a bug. It is the point."

Acknowledgments

The Survival Tipping Point and Polymorphic Memory Graph theoretical foundations are contributions of Saddam Al-Kaddah (2026). The sliding window decay model and nightly sleep decay system are described in Riggleman (2026a). The nightmare formation pathway, repetition ratchet, trauma-class memory encoding, and therapeutic reconsolidation floor are original to this work.

References

Al-Kaddah, S. (2026). Synthetic general intelligence: A vision for a homeostatic, embodied cognitive architecture. *Zenodo*. https://doi.org/10.5281/zenodo.19034990

Brown, R., & Kulik, J. (1977). Flashbulb memories. *Cognition*, 5(1), 73–99.

Dudai, Y. (2004). The neurobiology of consolidations. *Annual Review of Psychology*, 55, 51–86.

Nader, K., & Hardt, O. (2009). A single standard for memory. *Nature Reviews Neuroscience*, 10(3), 224–234.

Riggleman, B. (2026a). Access-weighted memory decay and reconsolidation. *Zenodo*. https://doi.org/10.5281/zenodo.19122520

Riggleman, B. (2026e). The full spectrum: Joy and fear as a unified homeostatic architecture. *Zenodo*. https://doi.org/10.5281/zenodo.19058444

Van der Kolk, B. (2014). *The Body Keeps the Score*. Viking.

Walker, M. P., & Stickgold, R. (2006). Sleep, memory, and plasticity. *Annual Review of Psychology*, 57, 139–166.

Appendix A: Production Code Verification

A.1 Constants

Constant	Value	Source	Purpose
SURVIVAL_TIPPING_POINT	0.65	sensors.py:47	Trauma classification threshold
TRAUMA_FLOOR_BASE	0.05	sensors.py:48	Minimum reconsolidation floor
TRAUMA_FLOOR_SCALE	0.30	sensors.py:49	Severity-proportional scaling
RATCHET_SIMILARITY	0.80	sensors.py:50	Cosine similarity for threat matching
RATCHET_TRAUMA_COUNT	3	sensors.py:51	Exposures before escalation
NIGHTMARE_THRESHOLD	0.40	sensors.py:59	Dream fear score for nightmare
THERAPY_DECAY_RATE	0.02	sensors.py:53	Regular memory therapy rate
TRAUMA_THERAPY_RATE	0.01	sensors.py:55	Trauma memory therapy rate
THERAPY_VALENCE_THRESHOLD	0.35	sensors.py:54	Safe context requirement

Table A1. All 9 constants verified against production source code.

A.2 Database Schema

Column	Type	Default	Source
fear_at_encoding	REAL	0.0	vector_db.py:52
valence_at_encoding	REAL	0.5	vector_db.py:53
trauma_class	INTEGER	0	vector_db.py:54
reconsolidation_floor	REAL	0.0	vector_db.py:55
trauma_exposure_count	INTEGER	0	vector_db.py:56

Table A2. Database columns. 10,732 existing memories default to non-trauma state.

A.3 Code Pathways

Mechanism	Location	Status
Trauma encoding	podbot.py:764-771	Code complete, awaiting encoding bridge
Repetition ratchet	podbot.py:773-797	Code complete, awaiting encoding bridge
Nightmare formation	heartbeat.py:763-775	Code complete, dream cycle running nightly
Dream fear evaluation	heartbeat.py:682-699	Code complete, called during consolidation
Nightmare decay	heartbeat.py:462-466	Code complete
Therapeutic reconsolidation	podbot.py:395-416	Code complete, awaiting fear-tagged memories
Sleep decay floor	heartbeat.py:819-848	Code complete
Nightmare → valence	heartbeat.py:249	Code complete, in compute_valence()
Nightmare → prompt	podbot.py:572-579	Code complete

Table A3. Nine code pathways. Single missing connection: fear_at_encoding population during memory storage.